| Given:<br>Due: | Further Maths (Decision)<br>HW 21 | Name: |
|---|---|---|

# Chapter 6: Linear programming

**Q1.** A manufacturer of frozen yoghurt is going to exhibit at a trade fair. He will take two types of frozen yoghurt, Banana Blast and Strawberry Scream.

He will take a total of at least 1000 litres of yoghurt.

He wants at least 25% of the yoghurt to be Banana Blast. He also wants there to be at most half as much Banana Blast as Strawberry Scream.

Each litre of Banana Blast costs £3 to produce and each litre of Strawberry Scream costs £2 to produce. The manufacturer wants to minimise his costs.

Let *x* represent the number of litres of Banana Blast and *y* represent the number of litres of Strawberry Scream.

Formulate this as a linear programming problem, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

You should **not** attempt to solve the problem.

**(Total 6 marks)**

Whilst the objective function was found correctly on many occasions, the absence of the word 'minimise' meant that the first mark could not be awarded. The first constraint (based on a total of at least 1000 litres of yoghurt) was usually correct. The other two constraints were either dealt with very well by students or not attempted at all. Simplified inequalities were not always seen and, on occasion, coefficients were left as fractions rather than integers.

| Question Number | Scheme | Marks |
|---|---|---|
| | Minimise $C = 3x + 2y$ | B1 |
| | Subject to: | |
| | $x + y \geq 1000$ | B1 |
| | $\frac{1}{4}(x+y) \leq x$, simplifies to $y \leq 3x$ | M1 A1 |
| | $2x \leq y$ | M1 A1 |
| | $(x, y \geq 0)$ | |
| | | **6 marks** |

**Notes for Question**

1B1: CAO – expression correct and 'minimise'.
2B1: CAO
1M1: Correct method – must see $\frac{1}{4}(x+y) \blacksquare x$ where $\blacksquare$ is any inequality or =. The bracket must be present or implied by later working.
1A1: CAO – simplified – answer must have integer coefficients.
2M1: Correct method – one of $2x \blacksquare y$ or $x \blacksquare 2y$ where $\blacksquare$ is any inequality or =.
2A1: CAO – answer must have integer coefficient.

**Q2.** Class 8B has decided to sell apples and bananas at morning break this week to raise money for charity. The profit on each apple is 20p, the profit on each banana is 15p. They have done some market research and formed the following constraints.

- They will sell at most 800 items of fruit during the week.
- They will sell at least twice as many apples as bananas.
- They will sell between 50 and 100 bananas.Assuming they will sell all their fruit, formulate the above information as a linear programming problem, letting *a* represent the number of apples they sell and *b* represent the number of bananas they sell.

Write your constraints as inequalities.

(7)

Many candidates omitted 'maximise' here, other common errors were omitting the nonnegativity constraint on *a*, and getting the 2 on the wrong side of the second inequality. A number of candidates tried to combine several conditions into one inequality, a frequently seen one being $2a + b \leq 800$. A number of candidates wasted time by starting to solve the LP problem.

| Question Number | Scheme | Marks |
|---|---|---|
| | Maximise (P=) $0.2a + 0.15b$ or $20a + 15b$ **o.e.** | B1 B1 (2) |
| | Subject to<br>$a + b \leq 800$<br>$a \geq 2b$<br>$50 \leq b \leq 100$<br>$a \geq 0$ | B1<br>B2,1,0<br>B1<br>B1<br>(5)<br><br>**Total 7** |
| | **Notes:**<br>1B1: 'Maximise'<br>2B1: ratio of coefficients correct<br>3B1: cao<br>4B1: ratio of coefficients of $a$ and $b$ correct.<br>5B1: inequality correct way round i.e. □$a \geq$□ $b$<br>6B1: cao accept < – accept two separate inequalities here<br>7B1: cao<br>• Penalise < and > only once with last B mark earned<br>• Be generous on letters a, b, A, B, x, y etc and mixed, but remove last B mark earned if inconsistent or 3 letters in the ones marked. | |

**Q3.**

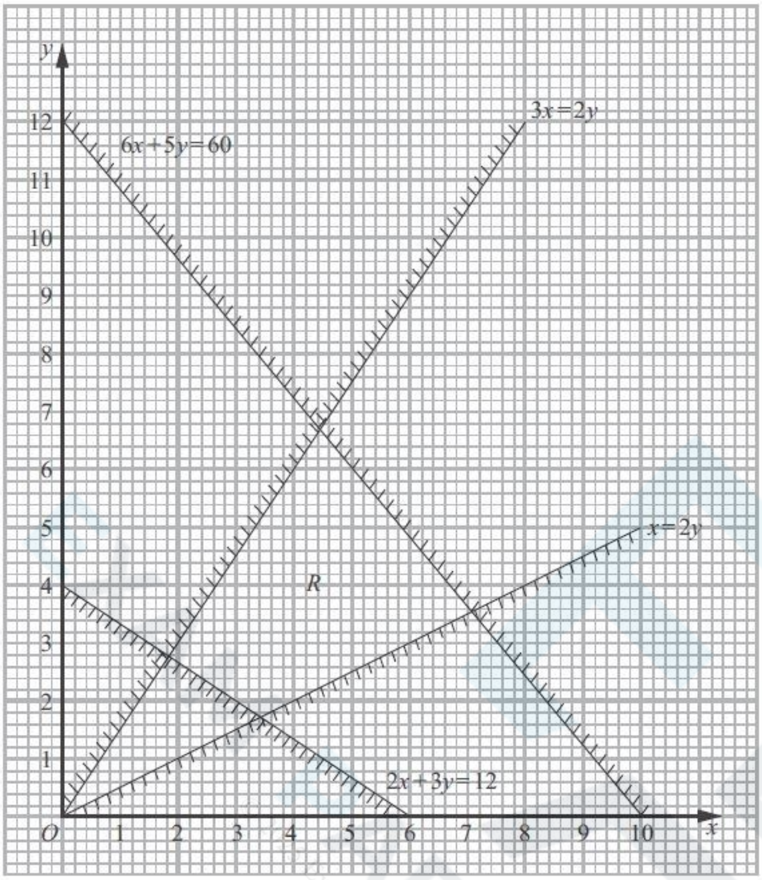



**Figure 2**

Figure 2 shows the constraints of a linear programming problem in $x$ and $y$, where $R$ is the feasible region.

(a) Write down the inequalities that form region $R$.

**(2)**

The objective is to maximise $3x + y$.

(b) Find the optimal values of $x$ and $y$. You must make your method clear.

**(4)**

(c) Obtain the optimal value of the objective function.

**(1)**

Given that integer values of $x$ and $y$ are now required,

(d) write down the optimal values of $x$ and $y$.

(1)

This question proved a good discriminator and differentiated the candidates well at all levels. In part (a) candidates generally did well selecting the correct inequality for the two line with positive gradients, but many got muddled dealing with the inequality of the lines with negative gradients. Those that used the objective line method in (b) were generally more successful, these candidates either stopped here or then went on to deal with the correct two lines leading to a solution that was either exact, or to 1 or more decimal places. Those that used point testing usually read directly from the graph and gave answers only to 1 decimal place. It was disturbing to see a good number of candidates spend time deriving all four points exactly and then not test any of them.

Part (c) was generally only done successfully by those who had found the point algebraically. A popular wrong answer for (d) was (7, 3) and many did not seem to understand the meaning of integer and restated their answer to (b).

| Question Number | Scheme | Marks |
|---|---|---|
| **(a)** | $6x+5y \le 60$<br>$2x+3y \ge 12$<br>$3x \ge 2y$<br>$x \le 2y$ | B2,1,0<br>(2) |
| **(b)** | Drawing objective line{ (0,3) (1,0)} Testing at least 2 points<br>Calculating optimal point Testing at least 3 points<br>$\left(7\frac{1}{17}, 3\frac{9}{17}\right)=\left(\frac{120}{17}, \frac{60}{17}\right) \approx (7.06, 3.53)$ | M1 A1<br>DM1<br>A1 awrt<br>(4) |
| **(c)** | $24\frac{12}{17}=\frac{240}{17} \approx 24.7$ (awrt) | B1<br>(1) |
| **(d)** | (6,4) | B1<br>(1)<br>**8** |
| **Notes:** | $\left(3\frac{3}{7}, 1\frac{5}{7}\right)=\left(\frac{24}{7}, \frac{12}{7}\right) \approx (3.43, 1.71) \rightarrow 12$<br>$\left(1\frac{11}{13}, 2\frac{10}{13}\right)=\left(\frac{24}{13}, \frac{36}{13}\right) \approx (1.85, 2.77) \rightarrow 8.307692 \left(8\frac{4}{13}=\frac{108}{13}\right)$<br>$\left(4\frac{4}{9}, 6\frac{2}{3}\right)=\left(\frac{40}{9}, \frac{20}{3}\right) \approx (4.44, 6.67) \rightarrow 20$<br>$\left(7\frac{1}{17}, 3\frac{9}{17}\right)=\left(\frac{120}{17}, \frac{60}{17}\right) \approx (7.06, 3.53) \rightarrow 24.705882 \left(24\frac{12}{17}=\frac{420}{17}\right)$ | |
| | **Notes** | |
| **(a)1B1** | Any two inequalities correct, accept < and > here (but not = of course). | |
| **2B1** | All four correct. Must be $\le$ and $\ge$ here | |
| **(b)1M1** | Drawing objective line or its reciprocal OR testing two vertices in the feasible region (see list above) points correct to 1 dp. | |
| **1A1** | Correct objective line OR two points correctly tested (1 dp ok) | |
| **2DM1** | Calculating optimal point either answer to 2 dp or better or using S.E's (correct 2 equations for their point + attempt to eliminate one variable.); OR Testing three points correctly and optimal one to 2dp. | |
| **2A1** | CAO 2 dp or better. | |
| **(c)B1** | CAO | |
| **(d)B1** | CAO not (4,6). | |

**Q4.** A linear programming problem in $x$ and $y$ is described as follows.

Maximise $P = 2x + 3y$

subject to

$$\begin{aligned} x &\geq 25 \\ y &\geq 25 \\ 7x + 8y &\leq 840 \\ 4y &\leq 5x \\ 5y &\geq 3x \\ x, y &\geq 0 \end{aligned}$$

(a) Add lines and shading to Diagram 1 in the answer book to represent these constraints. Hence determine the feasible region and label it R.

**(4)**

(b) Use the objective line method to find the optimal vertex, V, of the feasible region. You must clearly draw and label your objective line and the vertex V.

**(3)**

(c) Calculate the exact coordinates of vertex V.

**(2)**

Given that an integer solution is required,

(d) determine the optimal solution with integer coordinates. You must make your method clear.

**(2)**

Most students were able to draw the required lines correctly in question (a) although some were unable to draw lines sufficiently accurately (some drew lines without a ruler) or sufficiently long enough. The following general principle should always be adopted by students:

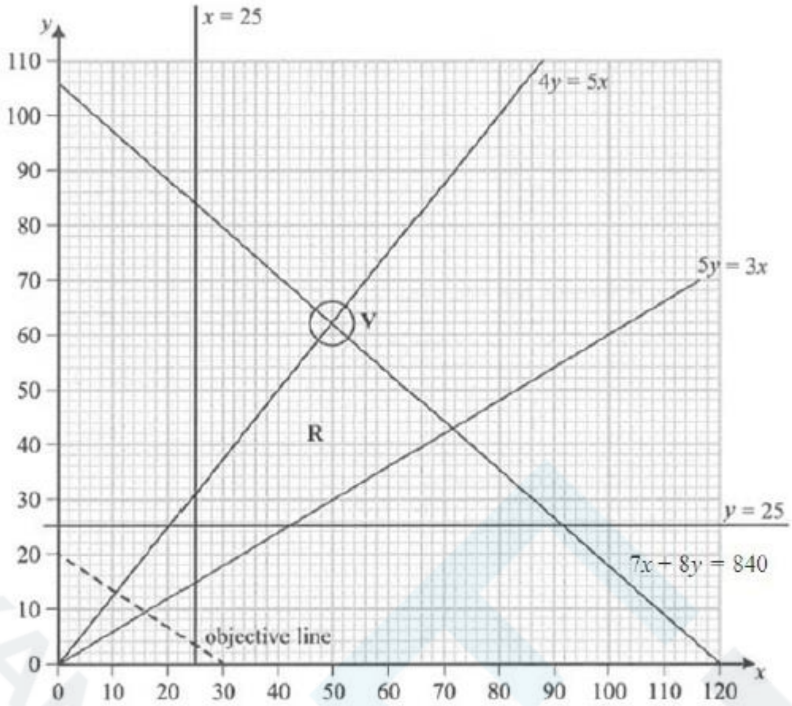
- lines should always be drawn which cover the entire graph paper supplied in the answer book and therefore,
- lines with negative gradients should always be drawn from axis to axis.

The rationale behind this is that until all the lines are drawn (and shaded accordingly) it is unclear which lines (or parts of lines) will define the boundary of the feasible region. If students only draw the line segments that they believe define the boundary of the feasible region then examiners are unaware of the order in which the lines were drawn and therefore it is unclear to examiners why some parts of the lines have been omitted. In general the lines $7x + 8y = 840$ , $x = 25$ and $y = 25$ were correctly drawn and where errors occurred they tended to be with the other two lines. Furthermore, a significant number of students were unable to select the correct feasible region.

In question (b) most students were able to draw a correct objective line, occasionally a line of reciprocal gradient was seen although this was fairly rare. It is worth noting that some students do not make their objective line clear and when it is drawn deep into the feasible region it can sometimes be difficult to identify. Students should be reminded to ensure that their objective line is labelled or distinct from their constraint lines. The majority of students successfully labelled the optimal vertex.

In question (c) some students did not demonstrate any working to find the exact coordinates of vertex V. Students are possibly relying on calculators to do this for them and these students need to be reminded of the advice to students on the front cover of the question paper that 'answers without working may not gain full credit'.

Question (d) was found to be quite discriminating. The majority of students did not test the integer points around their optimum vertex with the correct inequalities and often those who did attempt this testing did not demonstrate that sufficient testing had been undertaken.

| Question Number | Scheme | Marks |
|---|---|---|
| (a) |  | B1<br>B1<br>B1<br>B1 R **(4)** |
| **(b)** | Drawing an objective line accept reciprocal gradient<br>correct objective line minimum length equivalent to (0, 10) to (15,0)<br>V labelled correctly | M1<br>A1<br>A1 **(3)** |
| **(c)** | $V\left(49\frac{7}{17}, 61\frac{13}{17}\right)$ | M1 A1**(2)** |
| **(d)** | Testing the correct inequalities for points with integer coordinates<br>(50, 61) | M1<br>A1 **(2)**<br>**11 marks** |

| Notes for Question |
|---|
| In (a) lines **must** pass through one small square of the points stated:<br><br>$7x + 8y = 840$ passes through (0, 105), (40, 70), (80, 35), (120, 0)<br>$4y = 5x$ passes through (0, 0), (40, 50), (80, 100)<br>$5y = 3x$ passes through (0, 0), (50, 30), (100, 60)<br><br>a1B1: One line other than $x = 25$ or $y = 25$ correctly drawn.<br>a2B1: Two lines other than $x = 25$ or $y = 25$ correctly drawn.<br>a3B1: All five lines correctly drawn.<br>a4B1: Region, R, correctly labelled – not just implied by shading – must have scored all three previous marks in this part.<br>b1M1: Drawing the correct objective line or its reciprocal. Line must be correct to within one small square if extended from axis to axis.<br>b1A1: Correct objective line.<br>b2A1: V labelled clearly on their graph. This mark is dependent on the correct five line segments that define the boundary of the feasible region.<br>cM1: Simultaneous equation being used to find **their** V (but not from $x = 25$ or $y = 25$). Must get to $x = \ldots$ and $y = \ldots$<br>cA1: Correct coordinates of V stated **exactly** as $\left(\frac{840}{17}, \frac{1050}{17}\right)$ or $\left(49\frac{7}{17}, 61\frac{13}{17}\right)$. If the correct coordinates are stated exactly with no working then this scores M1A0.<br>d1M1: Testing the correct inequalities for at least three of (49, 61), (49, 62), (50, 61), (50, 62).<br>d1A1: CAO (50, 61). |

**Q5.**

Ben is a wedding planner. He needs to order flowers for the weddings that are taking place next month. The three types of flower he needs to order are roses, hydrangeas and peonies.

Based on his experience, Ben forms the following constraints on the number of each type of flower he will need to order.

- At least three-fifths of all the flowers must be roses.
- For every 2 hydrangeas there must be at most 3 peonies.
- The total number of flowers must be exactly 1000

The cost of each rose is £1, the cost of each hydrangea is £5 and the cost of each peony is £4

Ben wants to minimise the cost of the flowers.

Let $x$ represent the number of roses, let $y$ represent the number of hydrangeas and let $z$ represent the number of peonies that he will order.

(a) Formulate this as a linear programming problem in $x$ and $y$ only, stating the objective function and listing the constraints as simplified inequalities with integer coefficients.

**(7)**

Ben decides to order the minimum number of roses that satisfy his constraints.

(b) (i) Calculate the number of each type of flower that he will order to minimise the cost of the flowers.

(ii) Calculate the corresponding total cost of this order.

**(3)**

It was noted by examiners that this final question was attempted by most candidates and thus indicated that time pressure was unlikely to have been an issue for many. Some responses were incomplete but most of the time it seems that candidates completed as much as they were able to do.

This question gave rise to a mixed bag of responses. Despite involving three variables, the work required, at least initially in part (a), was standard and the question gave rise to the usual errors that have been observed over many previous sessions.

Almost all candidates worked with the provided variables $x$, $y$ and $z$. Some candidates initially began formulating the problem in terms of $r$, $p$ and $h$ although usually they switched to $x$, $y$ and $z$ and rewrote their work with the required variables. Those that didn't convert their variables usually petered out before getting very far with the question.

Most candidates stated the objective function correctly and many remembered to 'minimise' although this was certainly not universally the case. Many candidates were able to state the equality constraint for the total number of flowers although some candidates believed there were 100 flowers rather than the given 1000. As is usually the case, the inequality constraints were generally more problematic. Most candidates had success with the roses constraint and many immediately deduced that $x \geq 600$. Sometimes though, this inequality was in the wrong direction and less able candidates gave other incorrect interpretations including $3x \geq 5(y + z)$. By far the most challenging constraint was the hydrangea/peony constraint which was often incorrect. It was common to see the incorrect inequality $2y \geq 3z$ but also $3y \leq 2z$ or $2y \leq 3z$. Sometimes all the constraints were stated as equations with no inequalities and examiners also observed strict inequalities being used from time to time.

Often candidates did not seem to realise that they had been asked to eliminate $z$ and many simply worked with the constraints in all three variables. Of those that did attempt to eliminate $z$ however, some candidates set the hydrangea/peony constraint to be an equality and used this to eliminate $z$ rather than using the correct $x + y + z = 1000$ constraint. Some candidates who did make headway eliminating $z$ from the constraints, neglected to eliminate $z$ from the objective function.

Often candidates were able to pick up marks in part (b) despite incomplete responses to part (a). Most recognised the need for the total number of flowers to equal 1000 and correspondingly stated values to fit their constraints. Occasionally though, the minimum value of $x$ was not used as had been stipulated and quite often candidates did not state their solution in context despite being asked for the number of each type of flower. Indeed, some candidates seemed to become muddled with the variables and flowers and stated solutions that were not compatible with their constraints – seemingly mixing up hydrangeas and peonies. On several occasions, examiners saw responses where candidates had only stated the required values for $x$ and $y$ and omitted the value for $z$. Occasionally the total cost was not stated. Overall, the question performed well providing a wide range of marks and the opportunity for differentiation amongst the candidates.

| Question | Scheme | Marks | AOs |
|---|---|---|---|
| (a) | Minimise $(P =) x + 5y + 4z$ | B1 | 3.3 |
| | Subject to $x \geq \frac{3}{5}(x + y + z) (\Rightarrow 2x \geq 3y + 3z)$ | B1 | 3.3 |
| | $3y \geq 2z$ | B1 | 3.3 |
| | $x + y + z = 1000$ | B1 | 3.3 |
| | $z = 1000 - x - y$ substituted into objective and constraints gives | M1 | 3.1a |
| | Minimise $(P =) y - 3x (+ 4000)$ subject to | A1 | 1.1b |
| | $x \geq 600$ and $2x + 5y \geq 2000$ | A1 | 1.1b |
| | | (7) | |
| (b) | (i) Using least value of $x$ to find $y$ and $z$<br>600 roses, 160 hydrangeas and 240 peonies<br>(ii) £2360 | M1<br>A1<br>A1 | 3.4<br>3.2a<br>1.1b |
| | | (3) | |
| | | | (10 marks) |

| Notes |
| --- |
| **(a)**<br>**B1**: CAO (for objective) – must contain 'minimise' or 'min' only (so not 'minimum') either when stated in terms of $x$, $y$ and $z$ or $x$ and $y$ only<br>**B1**: $x \geq \frac{3}{5}(x+y+z)$ oe – need not be simplified for this mark, accept $x \geq \frac{3}{5}(1000)$<br>**B1**: $3y \geq 2z$ or any equivalent form (need not be simplified nor integer coefficients for this mark)<br>**B1**: $x+y+z=1000$ (could be implied by earlier/later working)<br>**M1**: Eliminating $z$ from either the objective **or** both constraints using the constraint $x+y+z=1000$<br>**A1**: Correct objective in terms of $x$ and $y$ only – condone lack of 'minimise'<br>**A1**: Both constraints correct ( $x \geq 600$ and $2x+5y \geq 2000$ - must be integer coefficients for this mark) |
| **(b)(i)**<br>**M1**: Using their least value of $x$ to find both $y$ and $z$ (with both $y$ and $z$ being positive integers) – note that all values must satisfy the constraint $x+y+z=1000$ (and must all be integers)<br>**A1**: All three types of flowers correct (in context – so not just in terms of $x$, $y$ and $z$) – must come from correct constraints in (a)<br>**(ii)**<br>**A1**: CAO for cost (condone lack of units but not 2360p) – must come from correct constraints in (a)<br><br>**SC for (b)** – for those candidates with the constraint $2y \geq 3z$ in (a) leading to 600 roses, 240 hydrangeas and 160 peonies (so not just in terms of $x$, $y$ and $z$) together with (£)2440 award **SC M1A1A0** in (b) |